Domain Theory and Abstract Interpretation: Verifying Neural Network Robustness via Galois Connections

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Abstract

Deep neural networks operate on continuous, high-dimensional spaces where exhaustive verification of algorithmic robustness is strictly computationally intractable. A formal semantics must therefore explain both why verifying these continuous equations is impossible via brute-force methods, and how a safe approximation can be selected as the intended verifiable meaning. This paper presents the order-theoretic answer to the state-space explosion problem in machine learning. Values and computational states are ordered by information content, specific bounding representations are utilized to represent infinite state spaces, and Galois connections combine compatible finite approximations to ensure safety guarantees. On a complete lattice, every continuous network operation has a safe abstract counterpart derived precisely through the composition of abstraction and concretization functions. After proving this foundational theorem, the paper applies it to two case studies: the formal definition of the interval domain, and an exploration of advanced relational domains like zonotopes that mitigate the over-approximation wrapping effect. The construction connects the core course topics of partially ordered sets, monotone maps, closure operators, and Galois connections directly to the modern challenge of artificial intelligence safety.

Contents

1 Introduction **3**

2 Mathematical Preliminaries **3**
2.1 Information as an Order 3
2.2 Lattices and Complete Lattices 4

3 Galois Connections and Closure Operators **4**

4 The Abstraction-Refinement Framework **5**
4.1 Deriving Abstract Semantics 5

5 The State-Space Problem in Neural Networks **6**

6 Case Study I: The Interval Domain **6**
6.1 Formalizing the Galois Connection 6
6.2 Deriving the Abstract ReLU Operation 7

7 Numerical Application: Forward Propagation of Intervals **7**
7.1 Step 1: Affine Transformation 8
7.2 Step 2: ReLU Activation 8
7.3 Step 3: The Output Layer 8
7.4 Verification Conclusion 9

8 Case Study II: Advanced Relational Domains **9**
8.1 The Wrapping Effect 9
8.2 Zonotopes and Polyhedra 9

9 Scope and Limitations **9**

10 Connection to the Course Material **10**

11 Conclusion **10**

1 Introduction

The deployment of deep neural networks, particularly within computer vision, medical imaging, and safety-critical autonomous systems, has demonstrated unprecedented empirical success. However, these models remain highly vulnerable to adversarial perturbations. An adversarial attack involves microscopic, often imperceptible alterations to input data that reliably induce catastrophic misclassification. Guaranteeing the safety and robustness of these models is no longer merely an engineering challenge; it is a fundamental mathematical requirement for their integration into reliable systems.

Proving that a neural network is robust against all possible adversarial attacks within a given perturbation radius is a highly complex task. From the perspective of computational complexity theory, exhaustively verifying the continuous activation space of a neural network is strictly intractable. Because the input space is continuous and the network transformations introduce high-dimensional non-linearities, the set of all possible concrete states—known as the concrete semantics—is infinite. The resulting state-space explosion dictates that empirical testing or naive reachability analysis will inherently fail to provide absolute safety guarantees.

To overcome this computational barrier and bridge the gap between empirical success and formal mathematical proof, formal verification relies on the framework of abstract interpretation. Originally formulated in the 1970s for static program analysis, abstract interpretation safely over-approximates infinite system behaviors using finite, computable representations.

This elegant solution is entirely formalized by order theory. By establishing a structural duality known as a Galois connection between partially ordered sets, it becomes possible to map the complex, infinite concrete state space of a neural network into a simplified, verifiable abstract domain. This paper explores the core lattice-theoretic foundations of Galois connections and demonstrates how they provide the strict mathematical bounds necessary to certify machine learning robustness.

2 Mathematical Preliminaries

Before exploring abstract interpretation, we must establish the foundational order-theoretic structures that allow us to compare different domains of information.

2.1 Information as an Order

Recall that a partially ordered set, or poset, is a pair $(P, \sqsubseteq)$ in which the relation $\sqsubseteq$ is reflexive, antisymmetric, and transitive. In the context of domain theory and abstract interpretation, the order is normally read as an *information order*:

$$x \sqsubseteq y$$

This notation indicates that y contains at least as much information as x, or alternatively, that y is a less precise (more general) bound than x. Thus, "larger" in the poset sense need not mean numerically larger. For spatial bounds in $\mathbb{R}^n$, $x \sqsubseteq y$ implies that the geometric region bounded by x is fully encompassed by the geometric region bounded by y.

2.2 Lattices and Complete Lattices

If every pair of elements in a poset P has a greatest lower bound (meet, $\sqcap$) and a least upper bound (join, $\sqcup$), then P is a lattice.

Definition 2.1. *A poset $(P, \sqsubseteq)$ is a **complete lattice** if every subset $S \subseteq P$ has both a supremum ($\bigsqcup S$) and an infimum ($\sqcap S$) in P.*

A complete lattice always possesses a global least element, $\bot = \bigsqcup \emptyset$, and a global greatest element, $\top = \sqcap \emptyset$. In a semantic model, $\bot$ can represent an unreachable state or an empty geometry, while $\top$ represents complete uncertainty (the entire state space).

3 Galois Connections and Closure Operators

To formalize the relationship between two different domains of computation—such as an infinite continuous space and a finite bounding box—we map values between them using order-preserving functions.

Definition 3.1. *Let P and Q be ordered sets. A map $f : P \to Q$ is **monotone** (or order-preserving) if $x \sqsubseteq_P y \implies f(x) \sqsubseteq_Q f(y)$.*

A Galois connection establishes a strict structural correspondence between two posets via a pair of monotone functions that map elements back and forth.

Definition 3.2. *Let $(C, \sqsubseteq_C)$ and $(A, \sqsubseteq_A)$ be partially ordered sets. A pair of monotone functions $\alpha : C \to A$ (the abstraction function) and $\gamma : A \to C$ (the concretization function) forms a **Galois connection** if for all $c \in C$ and $a \in A$:*

$$\alpha(c) \sqsubseteq_A a \iff c \sqsubseteq_C \gamma(a)$$

In this framework, α is referred to as the lower adjoint and γ as the upper adjoint. This biconditional inequality is the foundational rule that guarantees safety in machine learning verification. It asserts that abstracting a concrete set of states c and verifying it fits within an abstract bound a is mathematically identical to concretizing the abstract bound a and verifying that it completely envelops the original concrete states c.

Galois connections naturally induce closure operators, which are mathematically critical for establishing safe approximation bounds.

Theorem 3.1. *If $\alpha : C \to A$ and $\gamma : A \to C$ form a Galois connection, then the composite function $\gamma \circ \alpha : C \to C$ is a **closure operator** on C. Specifically, for all $x, y \in C$:*

1. $x \sqsubseteq_C \gamma(\alpha(x))$ *(Extensive)*
2. $x \sqsubseteq_C y \implies \gamma(\alpha(x)) \sqsubseteq_C \gamma(\alpha(y))$ *(Monotone)*
3. $\gamma(\alpha(\gamma(\alpha(x)))) = \gamma(\alpha(x))$ *(Idempotent)*

Proof. To prove extensivity, let $a = \alpha(x)$. Since $\sqsubseteq_A$ is reflexive, $\alpha(x) \sqsubseteq_A \alpha(x)$. By the definition of a Galois connection, substituting a into the equivalence yields $x \sqsubseteq_C \gamma(\alpha(x))$.

To prove monotonicity, assume $x \sqsubseteq_C y$. Since α is monotone, $\alpha(x) \sqsubseteq_A \alpha(y)$. Since γ is monotone, applying it to both sides yields $\gamma(\alpha(x)) \sqsubseteq_C \gamma(\alpha(y))$.

To prove idempotence, we first note that $\alpha(\gamma(a)) \sqsubseteq_A a$ for any $a \in A$ (by substituting $c = \gamma(a)$ and using $\gamma(a) \sqsubseteq_C \gamma(a)$). Applying this to $a = \alpha(x)$, we get $\alpha(\gamma(\alpha(x))) \sqsubseteq_A \alpha(x)$. Conversely, from extensivity $x \sqsubseteq_C \gamma(\alpha(x))$, applying the monotone α gives $\alpha(x) \sqsubseteq_A \alpha(\gamma(\alpha(x)))$. By antisymmetry, $\alpha(\gamma(\alpha(x))) = \alpha(x)$. Applying γ to both sides yields $\gamma(\alpha(\gamma(\alpha(x)))) = \gamma(\alpha(x))$. □

The extensive property ($x \sqsubseteq_C \gamma(\alpha(x))$) is the core mechanism of a *safe over-approximation*: if a set of neural network activations is mapped to the abstract domain and then mapped back to the concrete domain, the resulting mathematical region is guaranteed to encompass the original activations perfectly.

4 The Abstraction-Refinement Framework

In the context of computer science and formal verification, the pure mathematics of a Galois connection translates directly into the Abstraction-Refinement Framework formulated by Patrick and Radhia Cousot in 1977.

To verify a complex system, we define two distinct posets. The first is the **Concrete Domain**, denoted C. For a neural network layer with n neurons, the exact output is a continuous region within $\mathbb{R}^n$. Thus, the concrete domain is the powerset of $\mathbb{R}^n$, denoted $\wp(\mathbb{R}^n)$, ordered by standard set inclusion ($\subseteq$). This forms a complete lattice where joins are set unions and meets are set intersections.

The second is the **Abstract Domain**, denoted A. This poset represents a simplified, finite, and computable approximation space, ordered by an abstract inclusion operator $\sqsubseteq_A$.

4.1 Deriving Abstract Semantics

The most powerful application of a Galois connection is not merely proving that an abstraction is safe, but actually using it to *derive* the optimal abstract operations for the neural network.

If we have a concrete neural network operation $f : C \to C$ (such as an affine matrix transformation or a non-linear activation), the Galois connection provides an exact mathematical recipe to construct the most precise safe abstract operation, $f^\sharp : A \to A$. This is defined as:

$$f^\sharp = \alpha \circ f \circ \gamma$$

This formula dictates how the abstract domain must behave: take an abstract bound, concretize it into an infinite set of real numbers, apply the exact concrete network math to the entire set, and then abstract the result back into a new computational bound. Because $f^\sharp$ is constructed directly from the closure operator, it inherits the guarantee that it will never underestimate the reachable state space.

5 The State-Space Problem in Neural Networks

To understand why this lattice-theoretic construction is strictly necessary in machine learning, we must examine the continuous geometry of deep learning models.

Consider an image classification network. An input image can be represented as a vector $x_0 \in \mathbb{R}^n$. An adversarial attack attempts to find a visually identical image x' within a small perturbation radius ϵ, such that the network misclassifies x'. The set of all possible perturbed inputs forms a continuous, infinite region—typically an L_∞ norm ball centered at x_0.

As this region propagates through the network, it undergoes a series of operations:

1. **Affine Transformations:** $Z_i = W_i X_{i-1} + b_i$
2. **Non-linear Activations:** $X_i = \mathrm{ReLU}(Z_i)$

While affine transformations preserve the convexity of the initial L_∞ ball, non-linear activation functions distort the continuous input region into highly complex, non-convex polyhedra.

Verifying robustness requires computing the exact shape of this propagated region at the final output layer and proving it does not intersect with the decision boundary of an incorrect classification class. Because computing the exact reachable set through sequential non-linear operations scales exponentially with the number of neurons, exhaustive concrete evaluation is impossible.

6 Case Study I: The Interval Domain

To overcome the intractability of the concrete semantics, researchers utilize Galois connections to map the complex activation spaces into computable abstract domains. The simplest and most foundational of these is the **Interval Domain** (or Box Domain).

Rather than tracking an exact, highly irregular set of concrete activations $S \subseteq \mathbb{R}^n$, the interval domain bounds S within an n-dimensional hyperrectangle. Each dimension is defined independently by a lower and upper bound: $[l, u]$.

6.1 Formalizing the Galois Connection

Let $C = \langle \wp(\mathbb{R}), \subseteq \rangle$ be the concrete domain of activation values for a single neuron, and $A = \langle \mathbb{I}, \sqsubseteq_A \rangle$ be the abstract domain of intervals, where $[l_1, u_1] \sqsubseteq_A [l_2, u_2]$ if and only if $l_2 \leq l_1$ and $u_1 \leq u_2$.

The abstraction function $\alpha : C \to A$ maps a complex set of concrete activations to the tightest enclosing interval:

$$\alpha(S) = [\inf(S), \sup(S)]$$

The concretization function $\gamma : A \to C$ maps an interval back to the infinite set of concrete real numbers it contains:

$$\gamma([l, u]) = \{x \in \mathbb{R} \mid l \leq x \leq u\}$$

Theorem 6.1. *The pair (α, γ) forms a valid Galois connection between $\langle \wp(\mathbb{R}), \subseteq \rangle$ and $\langle \mathbb{I}, \sqsubseteq_A \rangle$.*

Proof. We must show that $\alpha(S) \sqsubseteq_A [l, u] \iff S \subseteq \gamma([l, u])$.

($\Longrightarrow$) Assume $\alpha(S) \sqsubseteq_A [l, u]$. By definition of α, this means $[\inf(S), \sup(S)] \sqsubseteq_A [l, u]$. By the established ordering on $\mathbb{I}$, it follows that $l \leq \inf(S)$ and $\sup(S) \leq u$. Therefore, for every $x \in S$, we have $l \leq x \leq u$. This implies $x \in \gamma([l, u])$, so $S \subseteq \gamma([l, u])$.

($\Longleftarrow$) Assume $S \subseteq \gamma([l, u])$. By definition of γ, for every $x \in S$, we have $l \leq x \leq u$. Because l is a lower bound for all elements in S and u is an upper bound, it must be that $l \leq \inf(S)$ and $\sup(S) \leq u$. This satisfies the condition for interval ordering, meaning $[\inf(S), \sup(S)] \sqsubseteq_A [l, u]$, which is exactly $\alpha(S) \sqsubseteq_A [l, u]$. □

6.2 Deriving the Abstract ReLU Operation

We now use the Galois connection derivation $f^\sharp = \alpha \circ f \circ \gamma$ to construct the abstract ReLU operation.

The concrete function applied to a set is $f(S) = \{\max(0, x) \mid x \in S\}$. Let our abstract interval be $a = [l, u]$.

1. **Concretize:** $\gamma([l, u]) = \{x \in \mathbb{R} \mid l \leq x \leq u\}$.
2. **Apply Concrete Transformation:** $f(\gamma([l, u])) = \{\max(0, x) \mid l \leq x \leq u\}$.
3. **Abstract the Result:** Because $\max(0, x)$ is a monotonically increasing function over the reals, the infimum of this set is strictly $\max(0, l)$ and the supremum is $\max(0, u)$. Applying α yields the new abstract interval:

$$\alpha(f(\gamma([l, u]))) = [\max(0, l), \max(0, u)]$$

Thus, the Galois connection explicitly proves that our abstract ReLU operation must be defined as $f^\sharp([l, u]) = [\max(0, l), \max(0, u)]$. No precision is lost in the 1-dimensional ReLU abstraction.

7 Numerical Application: Forward Propagation of Intervals

To demonstrate the interval abstraction in practice, consider a highly simplified neural network with a single hidden layer. The network takes a 2-dimensional input, applies an affine transformation followed by a ReLU activation, and outputs a 1-dimensional prediction.

Suppose our concrete input is the vector $x_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. An adversary is allowed to perturb any input feature by a maximum radius of $\epsilon = 0.1$. We wish to formally verify the safety property that the network's final output will always remain strictly positive (> 0), regardless of the perturbation.

Instead of testing infinite concrete points within the continuous ϵ-ball, we abstract the perturbed input into the interval domain. Our abstract input X_0 becomes a vector of intervals:

$$X_0 = \begin{bmatrix} [0.9, 1.1] \\ [1.9, 2.1] \end{bmatrix}$$

7.1 Step 1: Affine Transformation

Let the weights W_1 and biases b_1 for the hidden layer be defined as:

$$W_1 = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

We compute the abstract pre-activation state $Z_1 = W_1 X_0 + b_1$ using interval arithmetic. In interval arithmetic, the sum of two intervals is $[a, b]+[c, d] = [a+c, b+d]$, and multiplication by a scalar k scales the bounds appropriately (reversing them if k is negative).

$$Z_1 = \begin{bmatrix} 1 \cdot [0.9, 1.1] - 1 \cdot [1.9, 2.1] + 0 \\ 2 \cdot [0.9, 1.1] + 1 \cdot [1.9, 2.1] - 1 \end{bmatrix}$$

$$Z_1 = \begin{bmatrix} [0.9, 1.1] + [-2.1, -1.9] \\ [1.8, 2.2] + [1.9, 2.1] - 1 \end{bmatrix}$$

$$Z_1 = \begin{bmatrix} [-1.2, -0.8] \\ [2.7, 3.3] \end{bmatrix}$$

7.2 Step 2: ReLU Activation

Next, we apply the abstract ReLU activation function $f^\sharp([l, u]) = [\max(0, l), \max(0, u)]$ derived in Section 6.2 to obtain the post-activation state A_1:

$$A_1 = \begin{bmatrix} \max(0, [-1.2, -0.8]) \\ \max(0, [2.7, 3.3]) \end{bmatrix} = \begin{bmatrix} [0, 0] \\ [2.7, 3.3] \end{bmatrix}$$

Notice how the abstraction perfectly bounds the non-linearity. Any concrete value that fell into the first interval $[-1.2, -0.8]$ is guaranteed to be strictly negative, so the ReLU function safely collapses that entire dimension to exactly $[0, 0]$.

7.3 Step 3: The Output Layer

Let the weights and biases for the final output layer be $W_2 = \begin{bmatrix} 1 & 1 \end{bmatrix}$ and $b_2 = 0$. We compute the final abstract output Z_2:

$$Z_2 = W_2 A_1 + b_2 = 1 \cdot [0, 0] + 1 \cdot [2.7, 3.3] + 0 = [2.7, 3.3]$$

7.4 Verification Conclusion

The final abstract state tells us that for *any* possible adversarial perturbation within $\epsilon = 0.1$, the exact concrete output of the neural network is mathematically guaranteed to fall within the interval $[2.7, 3.3]$. Because the lower bound of this interval is 2.7, which is strictly greater than 0, we have successfully and formally verified our safety property without having to exhaustively compute the infinite concrete state space.

8 Case Study II: Advanced Relational Domains

While the Interval Domain provides strict mathematical bounds, it suffers heavily from what is known as the "Wrapping Effect". Because intervals track the minimum and maximum of each neuron independently, they completely ignore the relational correlations between neurons.

8.1 The Wrapping Effect

If Neuron A and Neuron B are highly correlated (e.g., $A = B$), the true concrete state space is a diagonal line in 2D space. However, if bounded by independent intervals $A \in [0, 1]$ and $B \in [0, 1]$, the abstract domain represents this as a full 1×1 square. When this square is passed through subsequent affine transformations, the volume of the over-approximation grows exponentially. Eventually, the abstract bounds become so loose that verification fails, not because the neural network is unsafe, but because the abstraction is too imprecise.

8.2 Zonotopes and Polyhedra

To combat over-approximation, more complex Galois connections have been established utilizing advanced abstract domains.

A **Zonotope** is a centrally symmetric convex polytope defined by a center vector and a set of error terms (generators). Unlike intervals, a zonotope can rotate and stretch to closely bound diagonal and highly correlated concrete geometries. The Galois connection maps the non-convex concrete activation space to the tightest enclosing zonotope.

An even more precise abstract domain is the **Polyhedral Domain**, which tracks arbitrary linear inequalities between neurons ($c_1x_1 + c_2x_2 \leq d$). This domain preserves nearly all affine relational information but incurs a massive computational cost, scaling at $O(2^n)$ in the worst case.

Modern verifiers, such as DeepPoly, utilize hybrid relational domains that strike a balance between the speed of the interval domain and the precision of the polyhedral domain, ensuring scalable yet precise verification.

9 Scope and Limitations

The Galois connection framework solves the existence and safety problem of verification, but it does not completely solve the precision problem.

As discussed regarding the wrapping effect, mapping complex geometries to simplified posets inherently involves over-approximation. A neural network may be perfectly robust in the concrete domain, but if the abstract domain accumulates too much error through successive layers, the final bounding box may intersect an unsafe region. When this happens, the verifier returns an "Unknown" result. Therefore, "the abstraction is safe" and "the semantics correctly bounds the exact execution without loss of precision" are related but distinct claims.

Furthermore, applying this framework to non-linear activations beyond ReLU—such as Sigmoid or Tanh—requires constructing significantly more complex abstract bounding functions. Bounding a Sigmoid function often requires Taylor series expansions or sophisticated tangent planes, increasing the computational overhead of the concretization and abstraction cycles.

10 Connection to the Course Material

The construction developed throughout this paper serves as an advanced applied extension of the course's core lattice theory concepts:

- **Partially Ordered Sets (Posets):** In abstract interpretation, posets represent the ordering of information content and spatial bounds.
- **Top and Bottom Elements ($\top, \bot$):** The bottom element represents an impossible concrete state or an empty geometry, while the top element represents total uncertainty.
- **Monotone Maps:** These ensure that expanding an input geometric bound cannot inadvertently shrink or reduce the output geometric bound, preventing safety violations.
- **Closure Operators:** Re-applying the abstraction-concretization loop yields an idempotent, extensive bounding box that safely over-approximates the truth.
- **Galois Connections:** Provide the formal bridge between the infinite concrete space and the finite abstract space, solving the computational intractability of verification.

Thus, abstract interpretation does not alter lattice and order theory. Rather, it selects the specific notions of completeness, monotonicity, and structure-preserving maps that perfectly match the behavioral requirements of safety-critical machine learning systems.

11 Conclusion

The mathematical elegance of domain and lattice theory extends far beyond pure algebra. Galois connections provide the exact structural rigor necessary to bridge continuous, infinite concrete machine learning models with discrete, computable abstract bounding domains.

By defining a safe, order-preserving mapping between these posets, abstract interpretation transforms the intractable problem of continuous state-space explosion into a finite, verifiable task. The interval domain and advanced relational domains demonstrate that this

framework is not merely an existence theorem; its mathematical mappings describe exactly how much bounding behavior has been justified at each layer of a deep neural network. The resulting abstract bound contains all exact concrete behavior and no arbitrary unverified information.

As deep learning architectures increasingly act as the foundation for safety-critical infrastructure, heuristic testing will no longer suffice. The absolute theoretical guarantees provided by posets, closure operators, and Galois connections will remain the fundamental mechanism for certifying algorithmic reliability.

AI Usage and Verification Statement

During the drafting of this research paper, large language models (ChatGPT/Gemini) were utilized to generate LaTeX formatting, summarize complex academic literature regarding neural network robustness, and structure the narrative flow of the mathematical proofs. All mathematical definitions, theorems, and lattice-theoretic proofs were manually verified against the primary course textbook (*Introduction to Lattices and Order* by Davey and Priestley). The integration of the interval domain case study and the numerical forward propagation was synthesized through independent derivation to ensure the theoretical accuracy of the Galois connection mappings.

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